3D Vectors & the Cross Product

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

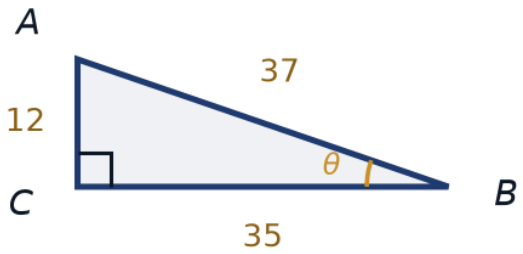
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the points $A(1, 2)$ and $B(4, 6)$. Find the vector $\vec{AB}$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use the compound angle formula for cosine to show that $\cos(180^\circ - x) = -\cos x,$ then find $\cos 150^\circ$. .



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-5;\ -2; 2)$ passing through $P(-3,\ -4,\ -3)$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Determine whether the lines $\mathbf{r}_1 = (1; 2;\ -1) + \lambda(2; 1; 3)$ and $\mathbf{r}_2 = (0; 5; 4) + \mu(-1; 2; 1)$ intersect or are skew.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Eliminate the parameter to determine the Cartesian equation of the curve $x = 1 + 2\sec t$, $y = -2 + 3\tan t$. State the shape, center, and vertices.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the angle θ between the vectors $\mathbf{u} = \langle 1, \sqrt{3} \rangle$ and $\mathbf{v} = \langle 2, 0 \rangle$ in degrees.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**
Level 1 · Foundations Evaluate $\iint_R (2x + y)\,dA$ where R is the rectangular region $[0, 3] \times [1, 2]$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**
Level 1 · Foundations Determine whether the vectors $\mathbf{u} = \langle 2, -3, 4 \rangle$ and $\mathbf{v} = \langle 6, 8, 3 \rangle$ are orthogonal, parallel, or neither.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**
Lines ℓ_1 and ℓ_2 have direction vectors $\mathbf{d_1} = (2; 1; -1)$ and $\mathbf{d_2} = (1; -1; 3)$. Find the acute angle between them.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**
[Multiple Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**
[Vectors & Vector-Valued Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**
[Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

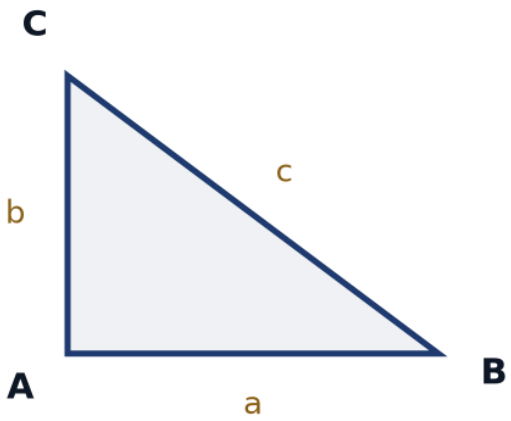
14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**
[Vectors, Lines & Parametric Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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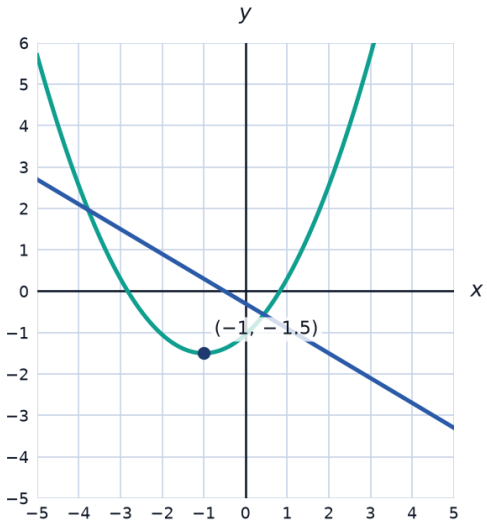
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.04 Partial Fractions

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.



16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

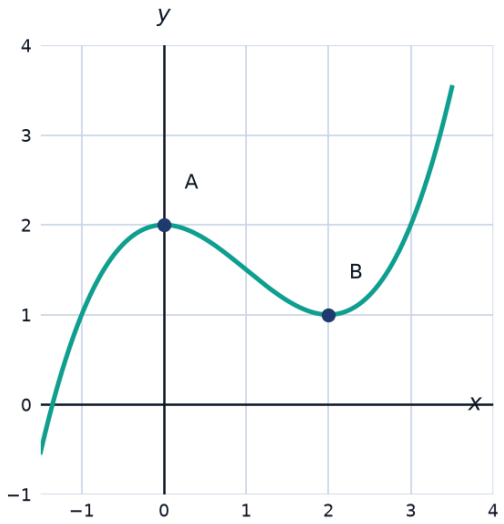
If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.



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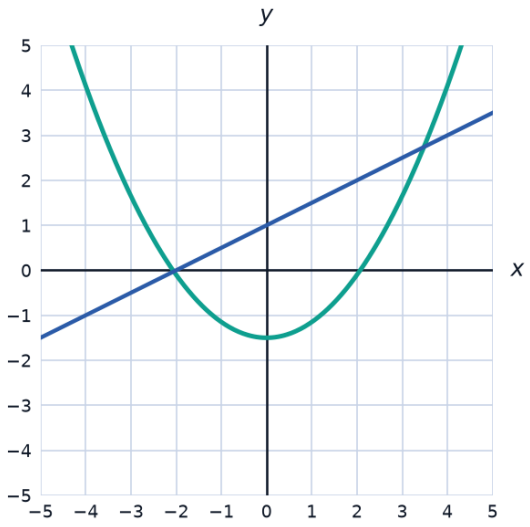
17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.



18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

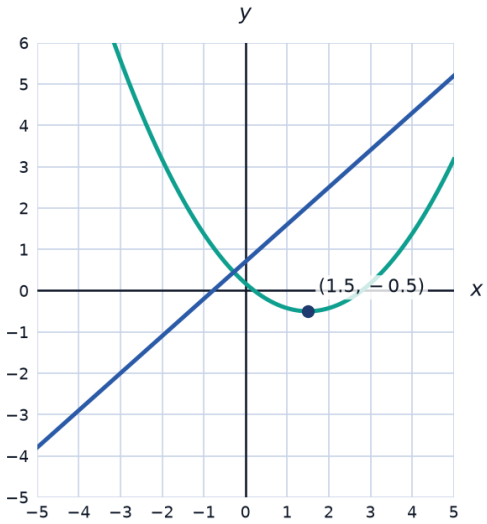
The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD 6*cmandtheradiusisUSD*3 cm, find the angle θ in radians.



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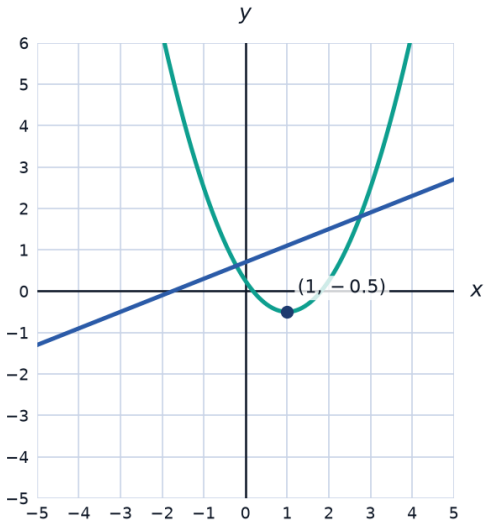
19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.



20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = [2, -3, 5]$ and $\mathbf{v} = [4, 1, -2]$.

a. -5

b. 5

c. 15

d. -1

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what value of k are the vectors $\mathbf{a} = [k, 3, -2]$ and $\mathbf{b} = [4, -2, 5]$ orthogonal (perpendicular)?

a. 4

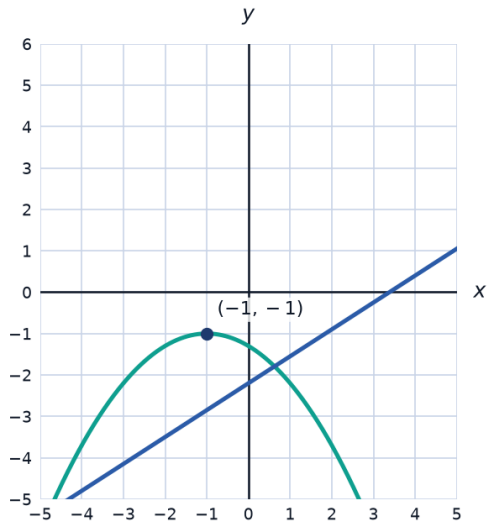
b. 2

c. -4

d. 5

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the angle θ between $\mathbf{u} = [1, 1, 0]$ and $\mathbf{v} = [0, 1, 1]$.



a. $\frac{\pi}{3}$ (60°)

b. $\frac{\pi}{4}$ (45°)

c. $\frac{\pi}{6}$ (30°)

d. $\frac{\pi}{2}$ (90°)

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Compute the cross product $\mathbf{u} \times \mathbf{v}$ where $\mathbf{u} = [1, 2, 3]$ and $\mathbf{v} = [4, 5, 6]$.

a. $[-3, 6, -3]$

b. $[3, -6, 3]$

c. $[-3, -6, -3]$

d. $[7, -2, -3]$

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the geometric relationship between the cross product vector $\mathbf{u} \times \mathbf{v}$ and the vectors $\mathbf{u}$ and $\mathbf{v}$?

a. $\mathbf{u} \times \mathbf{v}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$

b. $\mathbf{u} \times \mathbf{v}$ is parallel to $\mathbf{u} + \mathbf{v}$

c. $\mathbf{u} \times \mathbf{v}$ bisects the angle between $\mathbf{u}$ and $\mathbf{v}$

d. $\mathbf{u} \times \mathbf{v}$ lies in the plane spanned by $\mathbf{u}$ and $\mathbf{v}$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ for non-zero vectors $\mathbf{u}$ and $\mathbf{v}$, what does this indicate?

a. The vectors are parallel (collinear)

b. The vectors are perpendicular

c. The vectors have unit magnitude

d. The vectors are identical

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the area of the parallelogram spanned by vectors $\mathbf{a} = [2, 0, 1]$ and $\mathbf{b} = [1, 3, -2]$.

a. $\sqrt{59}$

b. $\sqrt{35}$

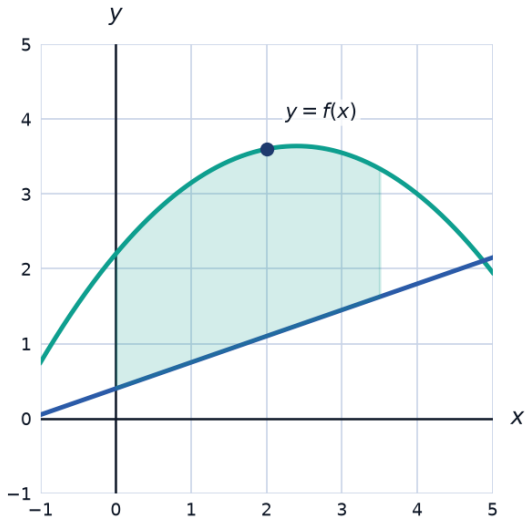
c. 14

d. $\frac{\sqrt{59}}{2}$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the area of the triangle with vertices $A(1, 0, 0), B(0, 2, 0), C(0, 0, 3)$.



a. $\frac{7}{2}$

b. 7

c. $\sqrt{14}$

d. 3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If two vectors have lengths $|\mathbf{u}| = 6$ and $|\mathbf{v}| = 5$, and the angle between them is 30°, what is $|\mathbf{u} \times \mathbf{v}|$?

a. 15

b. $15\sqrt{3}$

c. 30

d. $30\sqrt{3}$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What geometric quantity is given by the absolute value of the scalar triple product $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$?

a. The volume of the parallelepiped formed by $\mathbf{a}, \mathbf{b}, \mathbf{c}$

b. The surface area of the prism formed by $\mathbf{a}, \mathbf{b}, \mathbf{c}$

c. The length of the main diagonal vector

d. The perimeter of the base

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Calculate the volume of the parallelepiped with edges $\mathbf{a} = [1, 0, 0]$, $\mathbf{b} = [0, 2, 0]$, $\mathbf{c} = [1, 1, 3]$.

a. 6

b. 3

c. 12

d. 2

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$, what does this reveal about the three vectors?

a. The three vectors are coplanar (lie in the same plane)

b. The three vectors are mutually perpendicular

c. At least one vector must be the zero vector

d. The vectors form an equilateral triangle

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Using the compound angle formula, expand $\sin(x + 90^\circ)$.

a. $\cos x$

b. $-\cos x$

c. $\sin x$

d. $-\sin x$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A line has direction $\mathbf{d} = (2;\ -1;\ 0)$ and a plane has normal $\mathbf{n} = (1; 2; 2)$. Find the angle between the line and the plane.

a. 30°

b. 45°

c. 0°

d. 90°

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Line 1: $\mathbf{r} = (-3; 2;\ -3) + s(0; 0;\ -3)$. Line 2: $\mathbf{r} = (0;\ -1; 15) + t(-1; 1;\ -4)$. Find their point of intersection.

a. $(-5, 1, 5)$

b. $(-3, 2, 3)$

c. $(-4, 0, 4)$

d. $(-1, 1, 4)$

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Line: $\mathbf{r} = (2;\ -3; 0) + t(-3; 0; 2)$. Plane: $-x - 3y - 3z = 16$. Find the value of t at the point of intersection.

a. -4

b. -5

c. 0

d. -3

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the distance from $P(25, 11, 16)$ to the plane $6x + 2y + 3z = -25$.

a. 24

b. 35

c. 25

d. 43

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Line 1: $\mathbf{r} = (-2;\ -4; 2) + s(-2; 3;\ -1)$. Line 2: $\mathbf{r} = (0;\ -1; 4) + t(-4; 6;\ -2)$. Are the lines parallel?

a. Yes

b. No

c. They are the same line

d. They intersect at one point

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Find the distance from the origin to the plane $3x + 4y - 12z = 52$.

a. 5

b. 6

c. 4

d. 2

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Given $\tan\theta = 3/4$ with θ in Quadrant I, find $\csc\theta$.

a. $5/3$

b. $4/3$

c. $5/4$

d. $3/5$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.02 The Poisson Distribution

Points $A(-5,\ -1,\ -4)$ and $B(-1, 6,\ -3)$ are given. Find $\vec{AB}$.

a. $(6; 6; 3)$

b. $(5; 5; 0)$

c. $(4; 7; 1)$

d. $(2; 6; 0)$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Further Integration Techniques

Find the angle between lines with direction vectors $\mathbf{d_1} = (1; 4; 8)$ and $\mathbf{d_2} = (4;\ -1; 0)$.

a. 90°

b. 45°

c. 60°

d. 30°

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.02 Differential Equations

Find the exact value of $\cos 15^\circ$ using $\cos(45^\circ - 30^\circ)$.

a. $6 + 24$

b. $6 - 24$

c. $2 + 34$

d. $6 + 22$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

The magnitude of cross product $|\mathbf{a} \times \mathbf{b}|$ equals the area of the parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$. For $|a| = 3, |b| = 4, \theta = \pi/6$:

a. 24

b. 12

c. 3

d. 6

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Cross product $\mathbf{a} \times \mathbf{b}$ is orthogonal to:

a. $\mathbf{a}$ only

b. $\mathbf{b}$ only

c. neither

d. both $\mathbf{a}$ and $\mathbf{b}$

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